

Asset Pricing model

Approaches:

- consumption based model
- CAPM
- Arbitrage pricing Theory
- Multifactor model
- Option Pricing model
- Term structure model

- Functions: → fundamental analysis
- determine if observed price is 'wrong' → mispriced → trading opportunity
 - determine price asset should be if no observed price

Absolute Pricing

price assets by reference to **exposure** to fundamental source of macroeconomic risks

macroeconomic variables → inflation
economic growth
interest rates

Examples:

- consumption based model
- general equilibrium model

Relative Pricing

considers value of asset given **price** of some other asset

- little info used on fundamental risk factors
- no need to know where price of other assets came from

Examples:

- Black-Scholes Option Pricing
- Arbitrage Pricing

General Asset Pricing Formula

$$P_t = E \left[\sum_{i=1}^{\infty} m_{t+i} x_{t+i} \right]$$

→ asset payoff
[state price deflator]
stochastic discount factor

P_t → consistent with discounted CF approach

m_{t+i} → differs between pricing models

Asset Liability modelling (ALM)

Liability benchmark <
 static → not linked to L performance
 dynamic → changes as underlying L change

STAGES:

1. specify **key objectives** aimed to be achieved
 - future funding level
 - future solvency level
 - future company contribution rates
 - level of risk prepared to take → mismatch
2. Agree on suitable **assumptions** to use
3. **Data** on L needs to be collected to carry out projections
4. consider overall **nature** of L
 - current funding level
 - maturity
 - CF projections under different scenarios
5. Analysis carried out to find how 'interest' would progress in the future if different **investment strategies** were taken
6. Assess different **asset mixes** in more detail to assess **risk + reward** of alternative considerations
7. **Summarise** results + present in **graphic** format

CHAPTER 19: ACTUARIAL TECHNIQUES

ALM provides info in extra dimensions:

- ① Time
 - ↳ project into future
- ② Probabilistic
 - ↳ estimate on range of likely outcomes
- ③ Asset mix
 - ↳ effect of changing investment strategy